

01 — Product Scope

01 — Product Scope

Revision: 1 **Last modified:** 2026-07-05T14:20:00Z **Status:** Draft
— consolidated from 00-product-scope-and-principles.md and
Volume 1 (v01-product/).

1. Product definition

HelixVPN is a **self-hostable overlay network with a privacy-VPN front end**.

It unifies four lineages:

Borrowed from	What HelixVPN takes
Cloudflare Tunnel + WARP	Connector dials out; MASQUE/QUIC transport
Tailscale / Headscale	Coordination plane, network map, push-don't-poll, ACL-compiled topology
Mullvad	Obfuscation + privacy bar (MASQUE, Shadowsocks, UoT, LWO, DAITA, multi-hop, kill-switch, no-logging)
NetBird	Sanity-check OSS shape (WG + management/signal server, self-hosted)

Founding constraint: a remote user must obtain full, policy-scoped access to **one or many internal/home/lab networks without any inbound port-forward**.

2. What HelixVPN is / is NOT

2.1 It IS

1. A WireGuard-crypto-core overlay; obfuscation is a pluggable layer **beneath** WG.

2. A three-role system: Connector, Gateway, Client.
3. Self-hostable by one person, scaling to HA multi-region with the same images.
4. Cross-platform on iOS, Android, Aurora OS, HarmonyOS NEXT, Windows, Linux, macOS, Web.
5. Event-driven, push-based control plane.
6. No-logging-by-construction — CI-enforced, not a toggle.

2.2 It is NOT

NOT	Boundary
A forked or re-rolled WireGuard crypto	Transport spec owns wire shape only; CI lint forbids touching WG crypto primitives.
A separate "QUIC protocol" alongside WG	Mullvad's QUIC mode is WG-over-MASQUE/HTTP-3.
A system VPN in a browser	Web = Console management + optional browser-scoped WASM MASQUE proxy.
A logging / lawful-intercept / DLP appliance	Control actions are audited; traffic never is.
A generic L7 API gateway / service mesh	L3 IP overlay + L4 port policy only.
A consumer "free VPN" ad-funded service	Default is self-hosted; managed SKU is optional later.

3. Roles and apps

3.1 Three roles

Role	Runs	Direction	Reuses
Connector	Inside a private network	Outbound-only	helix-core advertise/route mode
Gateway	Public VPS	Accepts dials	helix-core edge + helix-go control plane
Client	End-user device	Outbound-only	helix-core + helix-ui

3.2 Three app classes

App	Persona	Primary jobs	Cores used
Helix Access	End user	Connect, pick exit/network, toggle obfuscation, kill-switch, split-tunnel	helix-ui + helix-core + shim
Helix Connector	Network operator	Onboard network, advertise CIDRs, local ACLs, run headless	helix-core (advertise/route) + optional helix-ui
Helix Console	Admin	Tenants, users, devices, networks, routes, policies, audit, billing-optional	helix-ui admin flavor only — no tunnel core

All three build from one Flutter tree via `runHelixApp(flavor, home, capabilities)`.

4. "Two-way" and "multi-network"

- **Two-way:** both the network-side leg (Connector → Gateway) and the user-side leg (Client → Gateway) are outbound; the Gateway stitches and polices between them.
- **Multi-network:** 1 user → N joined private networks, each policy-scoped. This requires solving:
 1. Overlapping RFC1918 ranges (D4).
 2. Per-user ACL routing (default-deny).
 3. Split horizon (connectors/clients cannot reach ungranted networks).

5. Prior art and positioning

System	Self-host	Obfuscation	Multi-network	Cross-platform apps
Mullvad	✗	best	✗	
Tailscale	partial	✗		
NetBird		limited		
Cloudflare Tunnel+WARP	✗	MASQUE		
HelixVPN		full		8 platforms,

System	Self-host	Obfuscation	Multi-network	Cross-platform apps
				one codebase

6. Hosting, licensing, and commercial stance

- **Self-hosted / home-lab first.** A managed SKU may come later using the same images.
- **Licensing (D8)** is an open decision; recommendation is **L-C**: permissive (Apache-2.0/MIT) reusable cores (helix-core, helix-ui, helix-proto) + source-available (BSL-class) managed Console layer. Owner-gated before public release.
- **No-logging is invariant** regardless of hosting shape.

7. Seven non-negotiable principles

1. **Control plane and data plane are strictly separated** — fail-static.
2. **WireGuard is the crypto core; transports are pluggable beneath it.**
3. **Outbound-only from edges** — no inbound holes.
4. **Push, don't poll** — WatchNetworkMap streaming deltas.
5. **One core per concern, reused everywhere.**
6. **Self-hostable by one person, scalable to many gateways.**
7. **No-logging by construction** — CI schema-lint fails on durable flow tables.

8. Mullvad parity matrix

#	Feature	Implementation	Phase
F1	WireGuard-only crypto	helix-core WG (kernel fast path, boringtun fallback)	P0/P1
F2	QUIC obfuscation (MASQUE)	helix-transport CONNECT-UDP via quinn+h3; edge :443/udp	P0/P1
F3	CONNECT-IP	quinn+h3 CONNECT-IP path	P2
F4	Shadowsocks	helix-transport Shadowsocks wrap	P2
F5	UDP-over-TCP	helix-transport UoT	P2
F6	LWO	helix-transport LWO (XOR/padding)	P1

#	Feature	Implementation	Phase
F7	Automatic obfuscation	Client escalation ladder	P1
F8	Custom WG port / 443 / 53	Edge multi-listener + port-hopping	P1
F9	DAITA	Optional shaping above WG (maybe-not-style)	P2
F10	Multi-hop	Nested WG, control-plane orchestrated	P2
F11	Kill-switch	OS firewall rules driven by core state machine	P1
F12	Split tunneling	Per-route AllowedIPs + per-app rules	P1
F13	DNS leak protection	Core sets tunnel DNS; blocks plaintext :53 off-tunnel	P1
F14	No-logging	Ephemeral Redis presence; no durable table	P1
F15	Anonymous account	Anonymous device-token enrollment alongside OIDC	P1
F16	Post-quantum handshake	WG PQ PSK / ML-KEM; hybrid never PQ-only	P2
F17	Per-device management	Registry + Console; revoke <1s	P1

HelixVPN differentiators: X1 multi-network overlay, X2 outbound-only onboarding, X3 self-hostable Mullvad-class stack, X4 direct P2P + NAT traversal + relay, X5 8-platform reach.

9. Scope per phase

Phase	In scope	Out of scope
P0 Spike	G1-G6: plain-UDP slice, MASQUE through DPI, iOS NE memory, Go-vs-Rust edge, FFI, static-map reconcile	Product UIs, persistence, multi-tenancy, HA
P1 MVP	Go monolith, Postgres+RLS, WatchNetworkMap, Redis Streams, identity/PKI, policy compiler, Gin REST/WS/SSE, helixvpctl, Podman quadlets, auto-ladder,	Shadowsocks/UoT/DAITA/P2P/multi-hop/PQ, desktop Windows/macOS, HA, Aurora/HarmonyOS/WASM

Phase	In scope	Out of scope
	iOS/Android/Linux Access, Connector, Console (web)	
P2 Parity+Reach	Full transport set, DAITA, P2P/NAT/relay, multi-hop, PQ, Windows/macOS desktop, GitOps, HA/multi-region	Aurora/ HarmonyOS/ WASM, billing, audit, reproducible builds
P3 Extended Reach	HarmonyOS, Aurora, WASM proxy, billing-optional multi-tenant, third-party audit, reproducible builds	L7 inspection, lawful-intercept, ad-funded free tier

10. Open architectural decisions D1-D8

ID	Decision	Recommendation	Gate
D1	Primary obfuscating transport	MASQUE/QUIC (Hysteria2 as ladder rung)	Phase-0 G2
D2	Shared client-core language	Rust core + Flutter UI	Phase-0 G3 (make-or-break)
D3	Event bus	Redis Streams MVP → NATS Phase 2	Phase-2 HA work
D4	Subnet collision	ULA /48 + 4via6 (NAT fallback)	Phase-1 IPAM design
D5	Gateway edge language	Rust (quinn+h3)	Phase-0 G4 benchmark
D6	Transport topology	Asymmetric per-leg via one crate	Phase-1 coordinator design
D7	MVP ambition	Lean spike → 8-criteria MVP	Settled
D8	Licensing model	L-C (permissive reusable cores + source-available managed layer)	Owner-gated before public release

11. Cross-references

- Functional requirements → [../v01-product/functional-requirements.md](#)

- Non-functional requirements → [../v01-product/nonfunctional-requirements.md](#)
 - Personas and RBAC → [../v01-product/personas-and-roles.md](#)
 - System architecture → [02 — System Architecture](#)
 - Data plane → [03 — Data Plane](#)
 - Control plane → [04 — Control Plane](#)
-

Sources: docs/research/mvp/final/00-product-scope-and-principles.md §1-§13, v01-product/functional-requirements.md, v01-product/nonfunctional-requirements.md, v01-product/personas-and-roles.md.